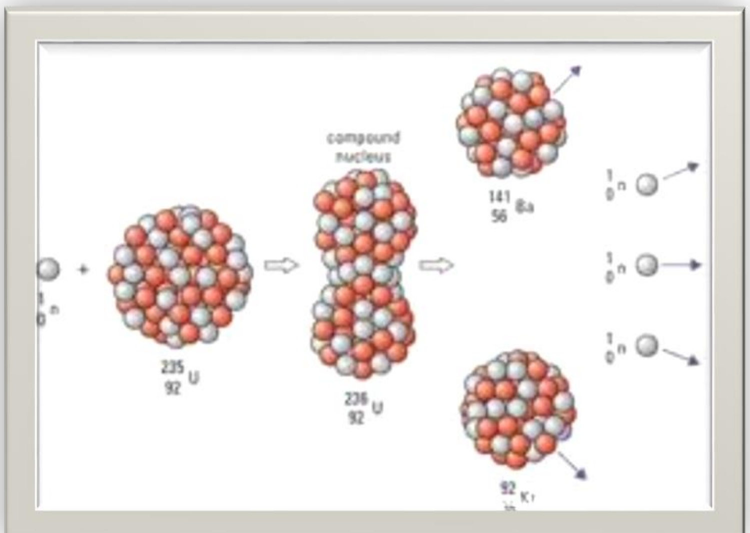
CHAPTER 13:
Nuclear reaction



Reaction Energy

- $Q = \Delta mc^2$
- $\Delta m = m_i - m_f$, m_i : mass of nucleus before reaction, m_f : mass of nucleus after reaction

  per nucleus (plural: nuclei)
- $Q = [(M_a + M_x) - (M_b + M_y)]c^2$ => mass x speed
- $Q = [(K_b + K_y) - (K_a + K_x)]$ => kinetic energy
- if $\Delta m / Q$ is (+) => exothermic, energy released
- if $\Delta m / Q$ is (-) => endothermic, energy absorbed
- $1u = 6.02 \times 10^{23}$ nuclei $= 1.6 \times 10^{-27}$ kg

Nuclear fission

- nuclear reaction which a heavier nucleus spilt into two lighter nuclei with almost same mass with emission of neutron and energy.

Nuclear fusion

- nuclear reaction which two lighter nuclei fuse to form a heavier nucleus and release large amount of heat at the same time

Lesson outlines:

13-1 Nuclear Reactions and the
Transmutation of Elements

13-2 Nuclear Fission; Nuclear Reactors

13-3 Nuclear Fussion

13-1 Nuclear Reactions and the Transmutation of Elements

A nuclear reaction takes place when a nucleus is struck by another nucleus or particle.

If the original nucleus is transformed into another, this is called transmutation.

An example:

$$^{4}_{2}\mathrm{He} + {}^{14}_{7}\mathrm{N} \rightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{H}.$$

13-1 Nuclear Reactions and the Transmutation of Elements

Energy and momentum must be conserved in nuclear reactions.

Generic reaction: $$\mathrm{a} + \mathrm{X} \rightarrow \mathrm{Y} + \mathrm{b},$$

<u>Reaction energy, Q</u>

Energy is released (liberated) in a nuclear reaction in the form of **kinetic energy of the particle emitted**, the **kinetic energy of the daughter nucleus** and the **energy of the gamma-ray photon** that may accompany the reaction.

It can be calculated by using formula:

$$\boxed{Q = (\Delta m)c^2}$$

Speed of light in vacuum

$$\text{Mass defect} = \sum \begin{matrix}\text{mass of nucleus} \\ \text{before reaction}\end{matrix} - \sum \begin{matrix}\text{mass of nucleus} \\ \text{products after reaction}\end{matrix}$$

$$\boxed{\Delta m = m_{\mathrm{i}} - m_{\mathrm{f}}}$$

$$\boxed{Q = ((M_a + M_X) - (M_b + M_Y))c^2}$$

- If the value of Δm OR Q is **positive**, the reaction is called **exothermic (exoergic)** in which the **energy released** in the form the kinetic energy of the product.
- If the value of Δm OR Q is negative, the reaction is called **endothermic (endoergic)** in which the **energy need to be absorbed** for the reaction occurred.

Because energy is conserved, Q has to be equal to the change in kinetic energy (final -initial):

$$\boxed{Q = (K_b + K_Y) - (K_a - K_X)}$$

13-1 Nuclear Reactions and the Transmutation of Elements

Example 1 :

Polonium nucleus decays by alpha emission to lead nucleus can be represented by the equation below:

$$^{212}_{84}\text{Po} \rightarrow ^{208}_{82}\text{Pb} + ^{4}_{2}\text{He} + Q$$

Calculate

a. the energy Q released in MeV.

b. the wavelength of the gamma-ray produced.

(Given mass of Po-212, m_{Po}=211.98885 u ; mass of Pb-208, m_{Pb}=207.97664 u and mass of α particle , m_{α}=4.0026 u)

$$Q = \frac{hc}{\lambda}$$

$$\lambda = \frac{6.67\times10^{-34}(3\times10^{8})}{8.95\times1.66\times10^{-19}}$$

$$= 1.4\times10^{-13}\ \text{m}$$

Solution :

(a) $\Delta m = m_i - m_f$

$= 211.98885 - (207.97664 + 4.0026)$

$= 9.61\times10^{-3}\ \text{u}$

$Q = \Delta m c^2$

$= \left(9.61\times10^{-3}\ \text{u} \times \frac{931.5\ \text{MeV}}{1\text{u}\cdot c^2}\right)c^2$

$= 8.95\ \text{MeV}$

13-1 Nuclear Reactions and the Transmutation of Elements

Solution :

$${}^{212}_{84}\text{Po} \rightarrow {}^{208}_{82}\text{Pb} + {}^{4}_{2}\text{He} + Q \quad \Rightarrow \quad \alpha \text{ decay}$$

before decay **after decay**

$$\underbrace{\sum Z_\text{i} = \sum Z_\text{f}} \quad \text{and} \quad \underbrace{\sum A_\text{i} = \sum A_\text{f}}$$

a. The mass defect (difference) of the reaction is given by

$$\begin{aligned}\Delta m &= m_\text{i} - m_\text{f} \\ &= m_\text{Po} - \left(m_\text{Pb} + m_\alpha\right) \\ &= 211.98885 - \left(207.97664 + 4.0026\right) \\ \Delta m &= 9.61 \times 10^{-3}\ \text{u}\end{aligned}$$

The energy released in the decay reaction can be calculated by using two method:

1st method:

$$\boxed{1\,\mathrm{u} = 1.66\times10^{-27}\ \mathrm{kg}}$$

$$Q = \underbrace{(\Delta m)}_{\text{in kg}}c^2$$

$$\Delta m = \left(9.61\times10^{-3}\right)\left(1.66\times10^{-27}\right)$$

$$= 1.5953\times10^{-29}\ \mathrm{kg}$$

$$Q = \left(1.5953\times10^{-29}\right)\left(3.00\times10^{8}\right)^2$$

$$Q = 1.436\times10^{-12}\ \mathrm{J}$$

$$Q = \frac{1.436\times10^{-12}}{1.60\times10^{-13}}$$

$$Q = 8.98\ \mathrm{MeV}$$

$$\boxed{1\,\mathrm{MeV} = 1.60\times10^{-13}\ \mathrm{J}}$$

2nd method:

$$\boxed{1\,\mathrm{u} = 931.5\ \mathrm{MeV}/c^2}$$

$$Q = \underbrace{(\Delta m)}_{\text{in u}}c^2$$

$$= \left[\Delta m\left(\frac{931.5\ \mathrm{MeV}/c^2}{1\,\mathrm{u}}\right)\right]c^2$$

$$= \left[\left(9.61\times10^{-3}\ \mathrm{u}\right)\left(\frac{931.5\ \mathrm{MeV}/c^2}{1\,\mathrm{u}}\right)\right]c^2$$

$$Q = 8.95\ \mathrm{MeV}$$

Solution :

b. The reaction energy Q is released in form of gamma-ray where its wavelength can be calculated by applying the Planck's quantum theory:

$$E = \frac{hc}{\lambda} = Q$$

$$\lambda = \frac{hc}{Q}$$

$$= \frac{\left(6.63\times10^{-34}\right)\left(3.00\times10^{8}\right)}{1.436\times10^{-12}}$$

$$\lambda = 1.39\times10^{-13}\ \text{m}$$

Note:

The **radioactive decay only occurred** when the value of Δm OR Q is **positive**.

Example 2: A slow neutron reaction.

$p = mv$

before momentum after momentum

The nuclear reaction $n + {}^{10}_{5}B \rightarrow {}^{7}_{3}Li + {}^{4}_{2}He$

p_i p_f

is observed to occur even when very slow-moving neutrons (mass M_n = 1.0087 u) strike a boron atom at rest.

$v_n = 0$

For a particular reaction in which $K_n \approx 0$, the outgoing helium (M_{He} = 4.0026 u) is observed to have a speed of 9.30 x 10^6 m/s. Determine :

(a) the kinetic energy of the lithium (M_{Li} = 7.0160 u)

(b) the Q-value of the reaction.

$$m_n v_n + m_B v_B = m_{Li} V_{Li} + m_{He} v_{He}$$

$$0 + 0 = 7.016(1.66\times10^{-27})V_{Li} + 4.0026(1.66\times10^{-27})(9.30\times10^{6})$$

$$0 = 1.16\times10^{-26} V_{Li} + 6.175\times10^{-20}$$

$$V_{Li} = -5.323\times10^{6}$$

$$K_E = \frac{1}{2} m_{Li} V_{Li}^2$$

$$= \frac{1}{2}(1.16\times10^{-26})(-5.323\times10^{6})^2$$

$$= 1.64\times10^{-13}\ J = 1.025\ MeV$$

$$Q = \Delta KE$$

$$= KE_f - KE_i$$

$$= (K_{Li} + K_{He}) - (K_B + K_e)$$

$$= 1.03 + 1.8 - 0$$

$$= 2.83\ MeV$$

$$K_{HE} = \frac{\frac{1}{2}(4.0026)(1.66\times10^{-27})(9.3\times10^{6})^2}{1.60\times10^{-13}}$$

$$= 1.7958\ MeV$$

13-1 Nuclear Reactions and the Transmutation of Elements

Example 3: Will the reaction "go"?

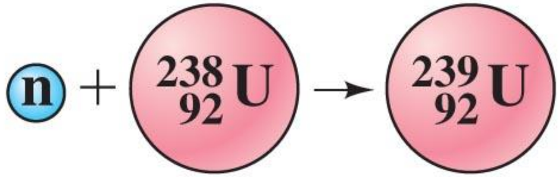
Can the reaction

$$p + {}^{13}_{6}C \rightarrow {}^{13}_{7}N + n$$

occur when ${}^{13}_{6}C$ is bombarded by 2.0-MeV protons?

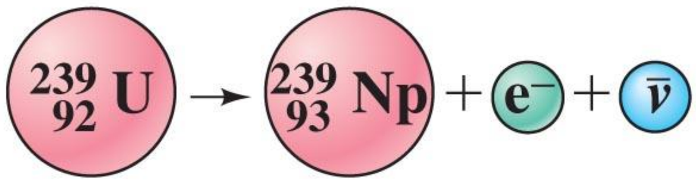
13-1 Nuclear Reactions and the Transmutation of Elements

(a)



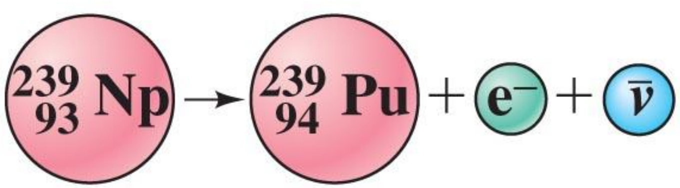
Neutron captured by $^{238}_{92}$U.

(b)



$^{239}_{92}$U decays by β decay to neptunium-239.

(c)



$^{239}_{93}$Np itself decays by β decay to produce plutonium-239.

Neutrons are very effective in nuclear reactions, as they have no charge and therefore are not repelled by the nucleus.

13-2 Nuclear Fission; Nuclear Reactors

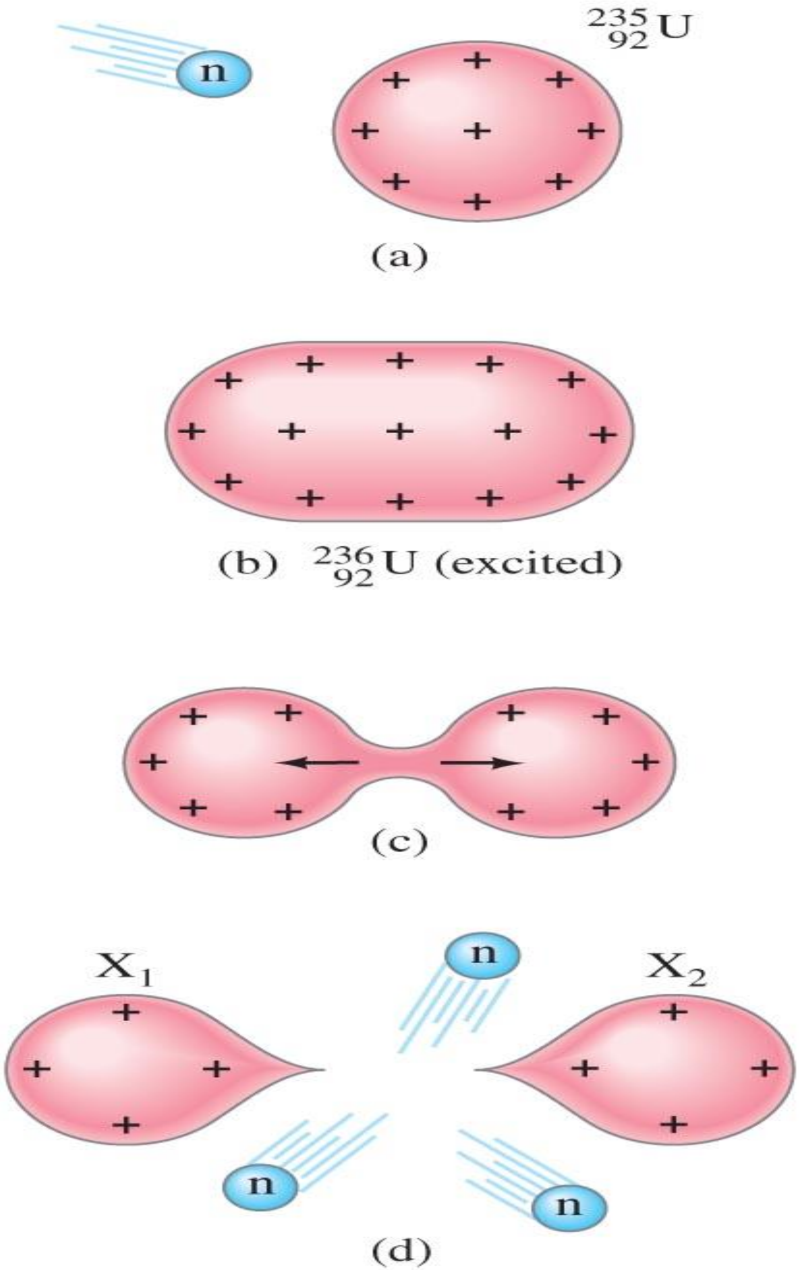
Nuclear fission

- is defined as **a nuclear reaction in which a heavy nucleus splits into two lighter nuclei that are almost equal in mass with the emission of neutrons and energy**.
- **Nuclear fission** releases an amount of energy that is **greater than** the energy released in **chemical reaction**.
- Energy is released because the **average binding energy per nucleon of the fission products is greater than that of the parent**.
- It can be divided into two types:
  - **spontaneous fission** – very rarely occur.
  - **induced fission** – bombarding a heavy nucleus with slow neutrons or thermal neutrons of low energy (about 10^{-2} eV). This type of fission is the important process in the energy production.

13-2 Nuclear Fission; Nuclear Reactors

After absorbing a neutron, a uranium-235 nucleus will split into two roughly equal parts.

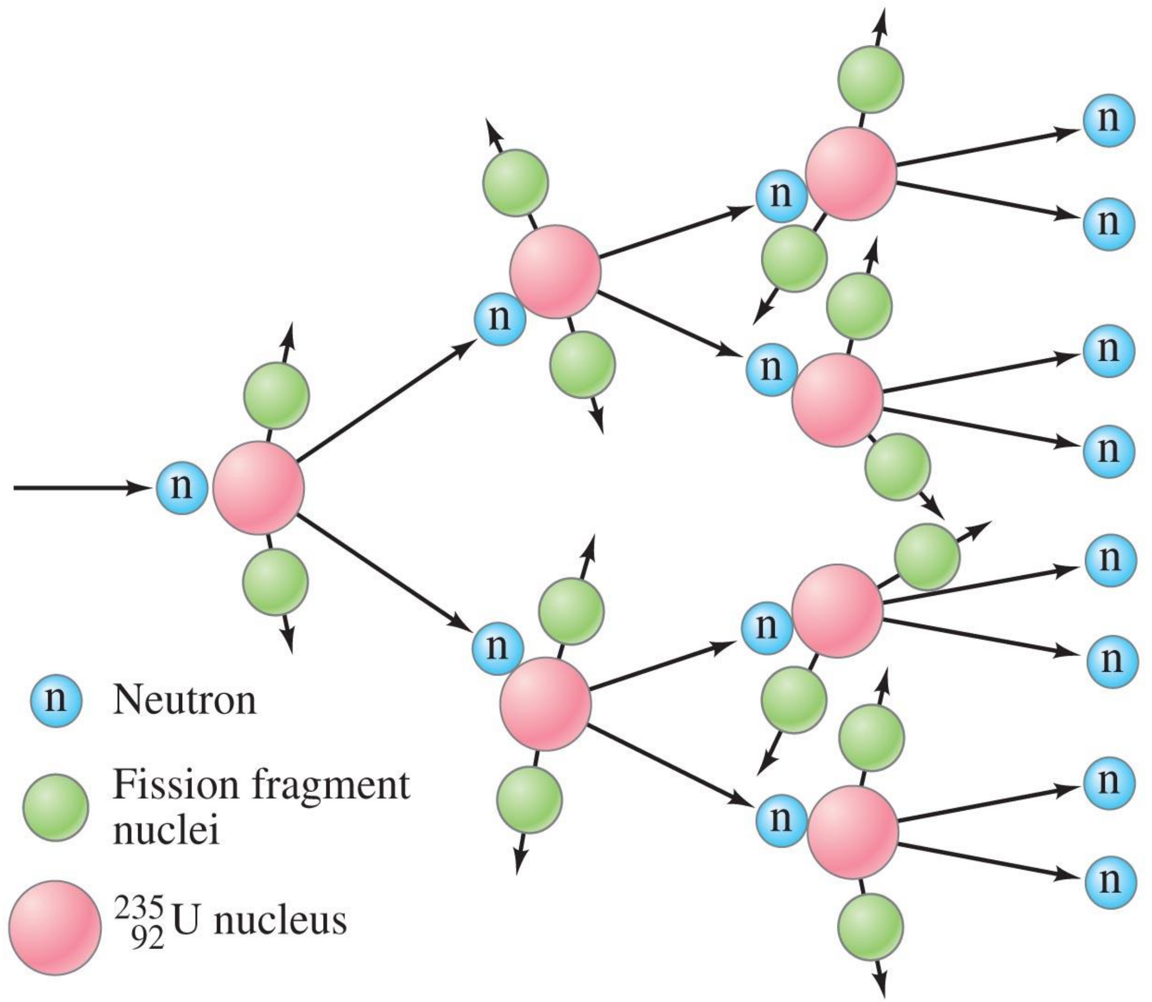
One way to visualize this is to view the nucleus as a kind of liquid drop.



13-2 Nuclear Fission; Nuclear Reactors

The energy release in a fission reaction is quite large. Also, since smaller nuclei are stable with fewer neutrons, several neutrons emerge from each fission as well.

These neutrons can be used to induce fission in other nuclei, causing a chain reaction.



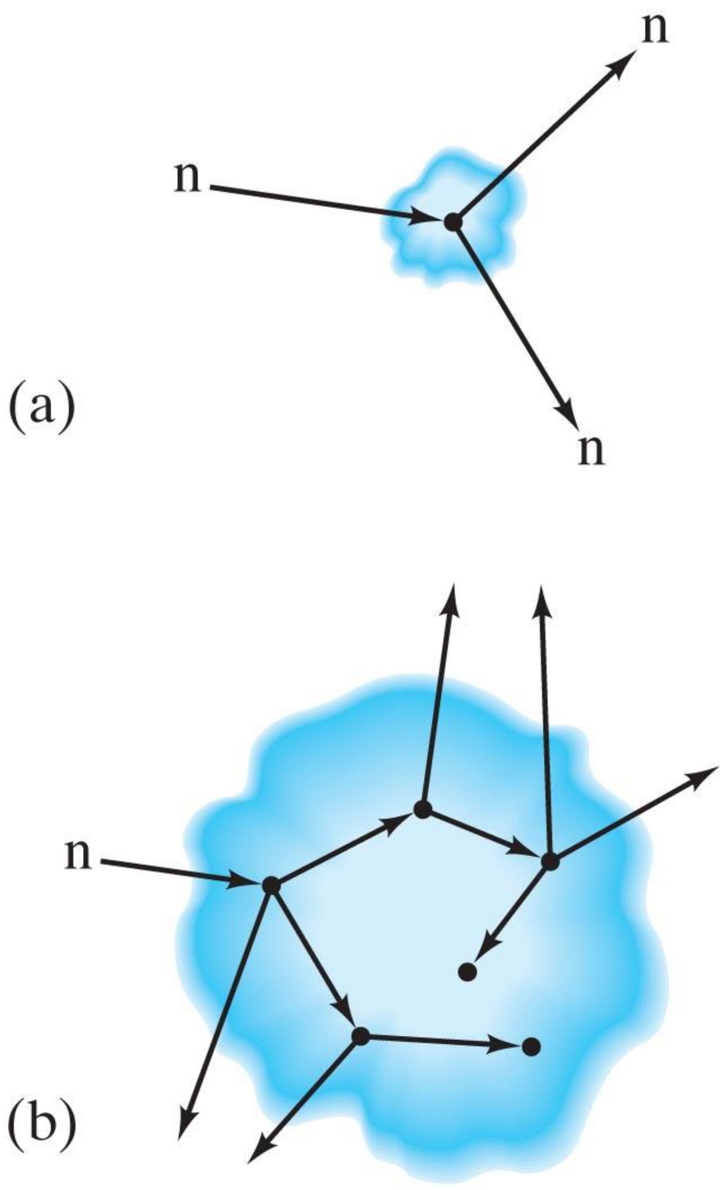
13-2 Nuclear Fission; Nuclear Reactors

In order to make a nuclear reactor, the chain reaction needs to be self-sustaining – so it continues indefinitely – but controlled.

A moderator is needed to slow the neutrons; otherwise their probability of interacting is too small. Common moderators are heavy water and graphite.

Unless the moderator is heavy water, the fraction of fissionable nuclei in natural uranium, about 0.7%, is too small to sustain a chain reaction. It needs to be enriched to about 2–3%.

13-2 Nuclear Fission; Nuclear Reactors



Neutrons that escape from the uranium do not contribute to fission. There is a critical mass below which a chain reaction will not occur because too many neutrons escape.

13-2 Nuclear Fission; Nuclear Reactors

The first use of fission, however, was not to produce electric power. Instead, it was first used as a fission bomb (called the "atomic bomb").

An **atomic bomb** also uses fission, but the core is deliberately designed to undergo a **massive uncontrolled chain reaction** when the uranium is formed into a critical mass during the detonation process.

Example 4 :

Calculate the energy released in MeV when 20 kg of uranium-235 undergoes fission according to

$$ {}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{89}_{36}\mathrm{Kr} + {}^{144}_{56}\mathrm{Ba} + 3\,{}^{1}_{0}\mathrm{n} + Q $$

(Given the mass of U-235 =235.04393 u, mass of neutron =1.00867 u, mass of Kr-89 =88.91756 u, mass of Ba-144 =143.92273 u and N_A =6.02×10^{23} mol^{-1})

Solution :

The mass defect (difference) of fission reaction for one nucleus U-235 is

$$
\begin{aligned}
\Delta m &= m_\mathrm{i} - m_\mathrm{f} \\
&= \left(m_\mathrm{U} + m_\mathrm{n}\right) - \left(m_\mathrm{Kr} + m_\mathrm{Ba} + 3m_\mathrm{n}\right) \\
&= \left(235.04393 + 1.00867\right) - \\
&\quad \left(88.91756 + 143.92273 + 3\times 1.00867\right)
\end{aligned}
$$

$$ \Delta m = 0.1863\mathrm{u} $$

Solution :

Q Δm one uranium

The energy released corresponds to the mass defect of one U-235 is

$$Q = (\Delta m)c^2$$

$$= \left[(0.1863\,\text{u})\left(\frac{931.5\,\text{MeV}/c^2}{1\,\text{u}}\right)\right]c^2$$

0.235 kg ⇓ 6.023×10^{23} nuclei

$$Q = 174\,\text{MeV}$$

one U-235

$(235 \times 10^{-3}$ kg of uranium-235$)$ contains of 6.02×10^{23} nuclei

20 kg of urainum-235 contains of $\left(\dfrac{20}{235 \times 10^{-3}}\right)\left(6.02 \times 10^{23}\right)$

$$= 5.12 \times 10^{25}\ \text{nuclei}$$

Therefore

Q_{total} = number of nuclei × Q

Energy released by 20 kg U-235 $= \left(5.12 \times 10^{25}\right)(174)$

$$= 8.91 \times 10^{27}\ \text{MeV}$$

13-3 Nuclear Fusion

Nuclear fusion

- is defined as **a type of nuclear reaction in which two light nuclei fuse to form a heavier nucleus with the release of large amounts of energy**.
- The **energy released** in this reaction is called **thermonuclear energy**.
- Examples of fusion reaction releases the energy are

$$ {}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + Q $$

$$ {}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} + Q $$

  - The two reacting nuclei in fusion reaction above themselves have to be brought into collision.

13-3 Nuclear Fusion

- As both nuclei are positively charged there is a strong repulsive force between them, which can only be overcome if the reacting nuclei have very high kinetic energies.
- These **high kinetic energies** imply temperatures of the order of **10^8 K**.

- At these elevated temperatures, however fusion reactions are **self sustaining** and the reactants are in form of a plasma (i.e. nuclei and free electron) with the nuclei possessing sufficient energy to overcome electrostatic repulsion forces.

- The **nuclear fusion reaction** can occur in fusion bomb and in the core of a star.

Comparison between fission and fusion

- Table 2 shows the **differences** between **fission** and **fusion** reaction.

Fission	Fusion
Splitting a heavy nucleus into two small nuclei.	**Combines** two small nuclei to form a larger nucleus.
It occurs at **temperature** can be **controlled**.	It occurs at **very high temperature** (10^8 K).
Easier to controlled and sustained.	**Difficult** to controlled and a sustained controlled reaction has not yet been achieved.

Table 2

- The **similarity** between the fission and fusion reactions is both reactions **produces energy**.
- Graph of binding energy per nucleon against the mass number in Figure 8 is used to explain the occurrence of fission and fusion reactions.

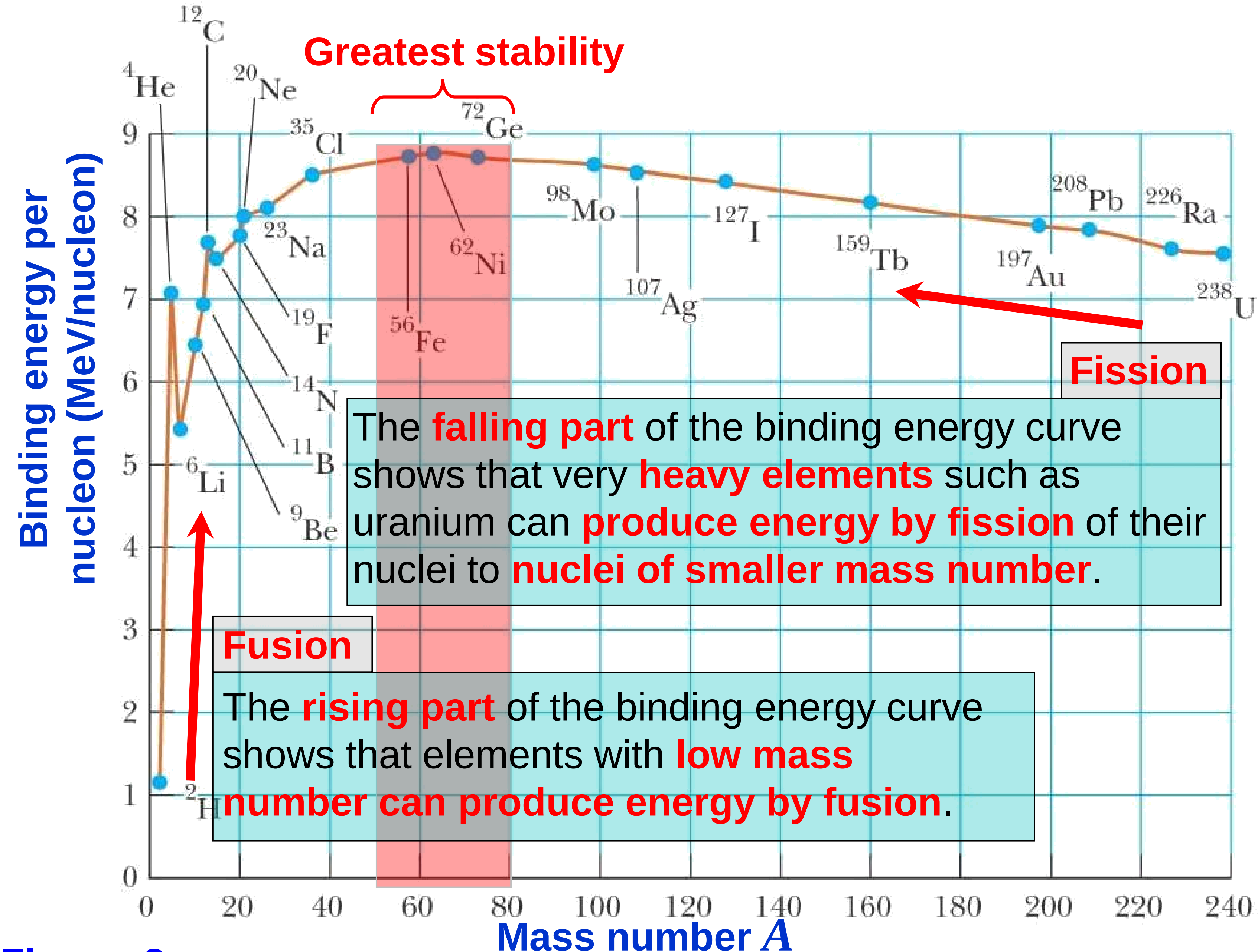

Greatest stability
Binding energy per nucleon (MeV/nucleon)
Mass number A
4He
12C
20Ne
35Cl
72Ge
23Na
62Ni
19F
56Fe
14N
11B
6Li
9Be
2H
98Mo
107Ag
127I
159Tb
197Au
208Pb
226Ra
238U
0
1
2
3
4
5
6
7
8
9
0
20
40
60
80
100
120
140
160
180
200
220
240
Fission
The falling part of the binding energy curve shows that very heavy elements such as uranium can produce energy by fission of their nuclei to nuclei of smaller mass number.
Fusion
The rising part of the binding energy curve shows that elements with low mass number can produce energy by fusion.

Figure 8

13-3 Nuclear Fusion

One of the simplest fusion reactions involves the production of **deuterium,** $^{2}_{1}H$, from a neutron and a proton:

$$^{1}_{1}H + n \longrightarrow {}^{2}_{1}H + \gamma.$$

13-3 Nuclear Fusion

The Sun creates power by fusing hydrogen into helium through the following set of reactions:

$$ {}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{2}_{1}\mathrm{H} + e^{+} + \nu \qquad (0.44\ \mathrm{MeV}) $$

$$ {}^{1}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + \gamma \qquad (5.48\ \mathrm{MeV}) $$

$$ {}^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \qquad (12.86\ \mathrm{MeV}) $$

13-3 Nuclear Fusion

The net effect is to transform four protons into a helium nucleus plus two positrons, two neutrinos, and two gamma rays.

$$4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2e^{+} + 2\nu + 2\gamma.$$

More massive stars can fuse heavier elements in their cores, all the way up to iron, the most stable nucleus.

Example 5 :

A fusion reaction is represented by the equation below:

$$ {}_{1}^{2}\mathrm{H} + {}_{1}^{2}\mathrm{H} \rightarrow {}_{1}^{3}\mathrm{H} + {}_{1}^{1}\mathrm{H} $$

Calculate

a. the energy in MeV released from this fusion reaction,

b. the energy released from fusion of 1.0 kg deuterium,

(Given mass of proton =1.007825 u, mass of tritium =3.016049 u and mass of deuterium =2.014102 u)

Solution :

a. The mass defect of the fusion reaction for 2 deuterium nuclei is

$$ \begin{aligned} \Delta m &= m_{\mathrm{i}} - m_{\mathrm{f}} \\ &= (m_{\mathrm{D}} + m_{\mathrm{D}}) - (m_{\mathrm{T}} + m_{\mathrm{p}}) \\ &= (2.014102 + 2.014102) - (3.016049 + 1.007825) \end{aligned} $$

$$ \Delta m = 4.33 \times 10^{-3}\ \mathrm{u} $$

Solution :

a. Therefore the energy released in MeV is

$$Q = (\Delta m)c^2$$

$$= \left[\left(4.33 \times 10^{-3}\ \text{u}\right)\left(\frac{931.5\ \text{MeV}/c^2}{1\ \text{u}}\right)\right]c^2$$

$$Q = 4.03\ \text{MeV}$$

b. The mass of 2 deuterium nuclei is 4×10^{-3} kg.

4×10^{-3} kg of deuterium contains of 6.02×10^{23} nuclei

1.0 kg of deuterium contains of $\left(\frac{1.0}{4 \times 10^{-3}}\right)\left(6.02 \times 10^{23}\right)$

$$= 1.505 \times 10^{26}\ \text{nuclei}$$

Therefore

Energy released from 1.0 kg deuterium $= \left(1.505 \times 10^{26}\right)(4.03)$

$$= 6.07 \times 10^{26}\ \text{MeV}$$

13-3 Nuclear Fusion

Example 6: Estimating fusion energy.

Estimate the energy released if the following reaction occurred:

$$^{2}_{1}H + ^{2}_{1}H \rightarrow ^{4}_{2}He.$$

$\Delta m = 2.014 + 2.014 - 4.0026 = 0.0254u$

$Q = \Delta mc^2$

$= 0.0254u \times 931.5$

$= 23.66 J$

13-3 Nuclear Fusion

There are three fusion reactions that are being considered for power reactors:

$$ {}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{1}H + {}^{1}_{1}H \qquad (4.00\,\mathrm{MeV}) $$

$$ {}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + n \qquad (3.23\,\mathrm{MeV}) $$

$$ {}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + n. \qquad (17.57\,\mathrm{MeV}) $$

These reactions use very common fuels – deuterium or tritium – and release much more energy per nucleon than fission does.